Predictive Modeling Using Machine Learning

1. Objective

This project aims to build a predictive machine learning model that can predict passenger survival using the Titanic dataset. The goal is to apply supervised learning techniques to analyze patterns in the data and evaluate the performance of different models.

2. Dataset Description

The dataset used for this project is the Titanic dataset, which contains information about passengers such as age, gender, ticket class, fare, and survival status. The dataset includes both numerical and categorical features that influence survival.

3. Data Preprocessing

The dataset was cleaned and prepared before applying machine learning algorithms. Missing values in the Age column were replaced with the mean value, and missing values in the Embarked column were filled with the most frequent value. The Cabin column was removed due to excessive missing values. Categorical variables such as Sex and Embarked were converted into numerical form for model training.

4. Feature Selection

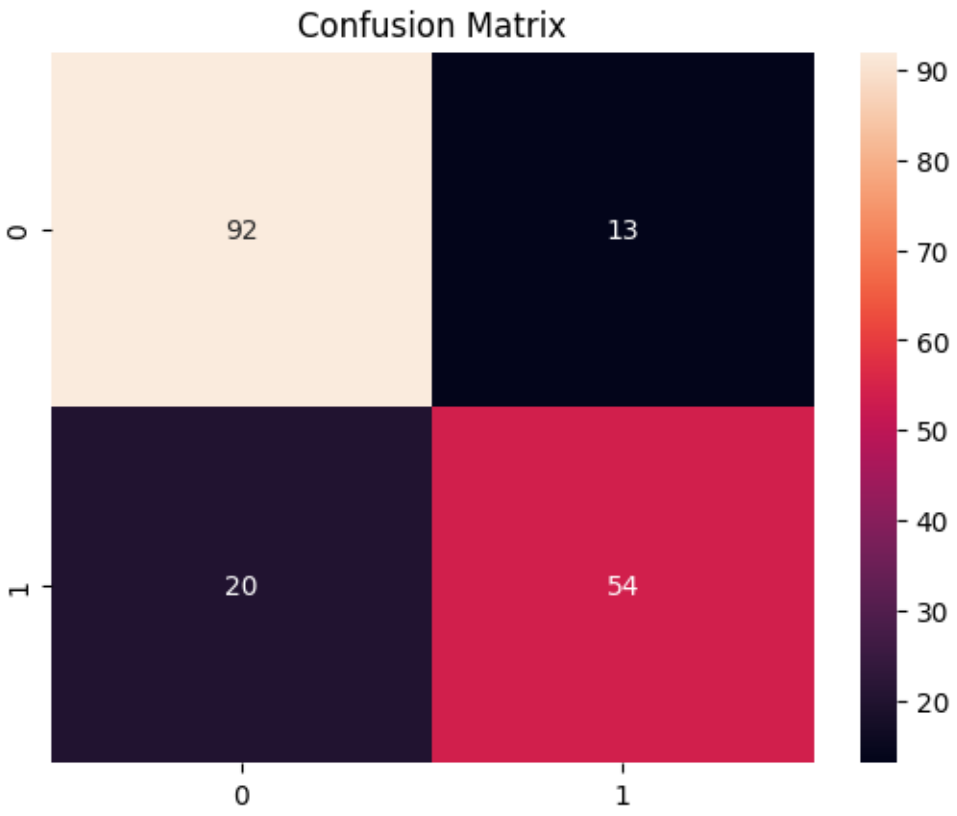
The following features were selected for building the predictive model: Age, Sex, Pclass, Fare, SibSp, Parch, and Embarked. The target variable used for prediction is Survived.

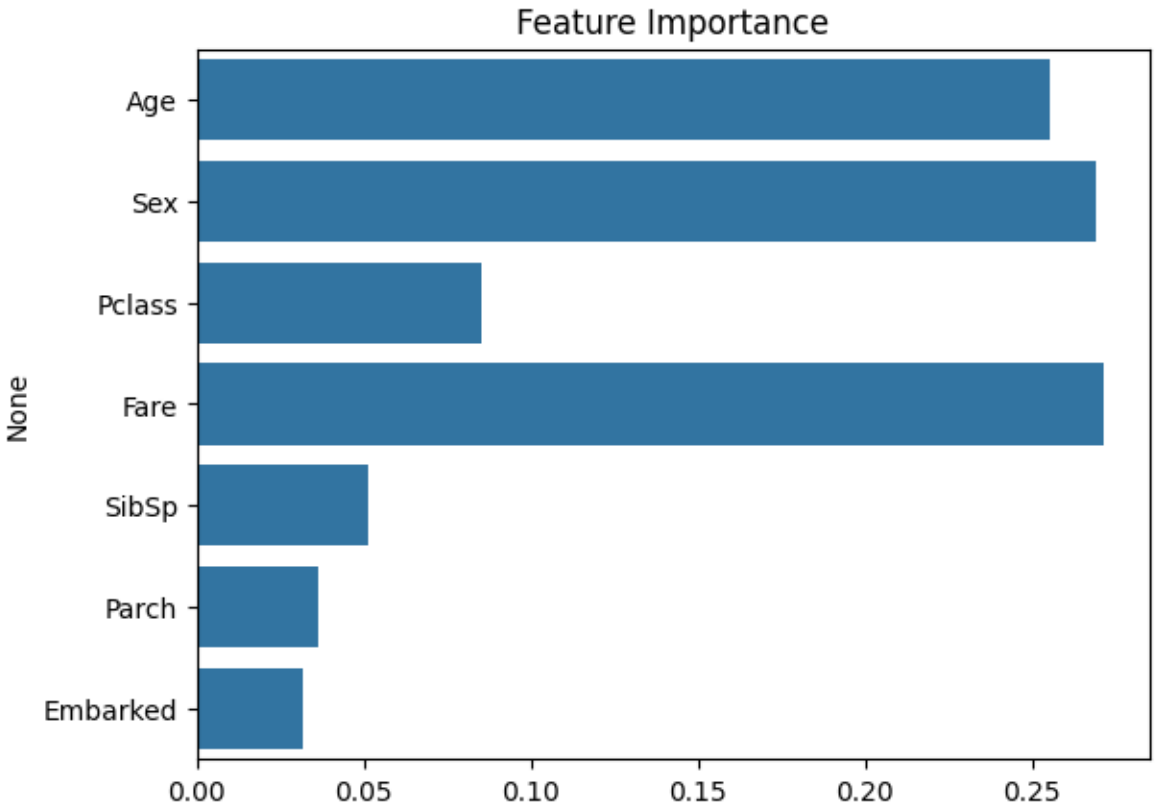
5. Machine Learning Algorithms Used

Three supervised learning algorithms were applied in this project: Linear Regression, Decision Tree Classifier, and Random Forest Classifier. These models were trained on the training dataset and tested on the testing dataset.

6. Model Evaluation

The models were evaluated using accuracy score, confusion matrix, and classification report. These metrics help measure how well the model predicts survival outcomes and how accurately it classifies passengers.


Confusion Matrix
92
13
20
54
0
1
0
1
90
80
70
60
50
40
30
20


Feature Importance
Age
Sex
Pclass
Fare
SibSp
Parch
Embarked
None
0.00
0.05
0.10
0.15
0.20
0.25

7. Results

Among the applied models, the Random Forest classifier achieved better prediction accuracy compared to the other models. This indicates that ensemble methods can provide stronger predictive performance for this dataset.

8. Conclusion

This project demonstrates how machine learning can be used to build predictive models using real-world data. The results show that features such as passenger class, gender, and fare significantly influence survival prediction. Overall, Random Forest proved to be the most effective model for this dataset.

9. Author

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